INCH-POUND

MS90546B
W/AMENDMENT 1
25 February 2021
SUPERSEDING
MS90546B
7 September 2007

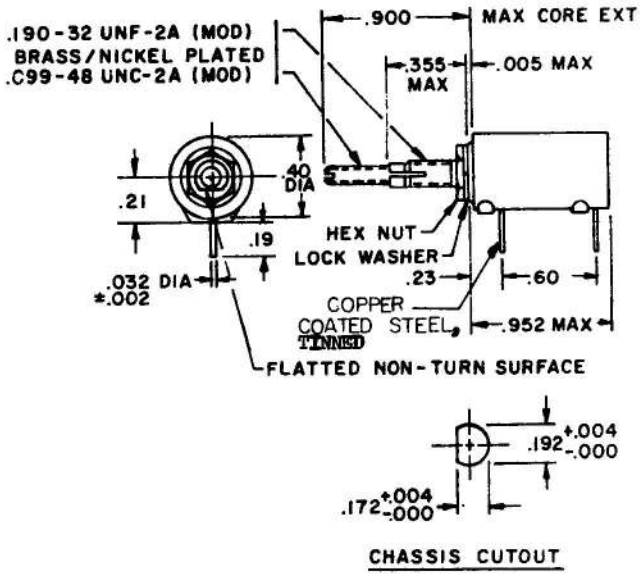
MILITARY SPECIFICATION SHEET

COILS, RADIO FREQUENCY, MOLDED, VARIABLE, FERRITE CORE (PRINTED CIRCUIT MOUNTING), TYPES LT12V032 TO LT12V062, INCLUSIVE

Inactive for new design after 31 March 1999

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.



INCHES	mm
.002	.05
.004	.10
.005	.13
.032	.81
.099	2.51
.172	4.37
.19	4.83
.192	4.88
.21	5.33
.23	5.84
.355	9.02
.40	10.16
.60	15.24
.900	22.86
.952	24.18

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are .XX ±.02 and .XXX ±.005.

FIGURE 1. Dimensions and configuration.

AMSC N/A

FSC 5950

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Style: LT12

Grade: 3
Class: A

Approximate weight:

Dash number 32 thru 42 - 5.8 grams.
43 thru 52 - 6.6 grams.
53 thru 62 - 7.0 grams.

Operating temperature range: -55°C to +105°C.

Ambient temperature: +90°C ±5°C.

Temperature rise: 15 °C, maximum.

Altitude: 70,000 feet.

Shock (specified pulse): MIL-STD-202-213, test condition I, is applicable. Coil shall be mounted by its mounting stud securely fastened with mounting hardware for vibration and shock testing.

Dielectric withstanding voltage (sea level): MIL-STD-202-301, test voltage 1000 V rms for a minimum of 60 seconds.

Barometric pressure (reduced): MIL-STD-202-105, test condition C, test voltage of 200 V rms for a minimum of 60 seconds.

Terminals: Terminals shall be in accordance with ASTM B227.

Terminal pull: 5 pounds, minimum.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I.

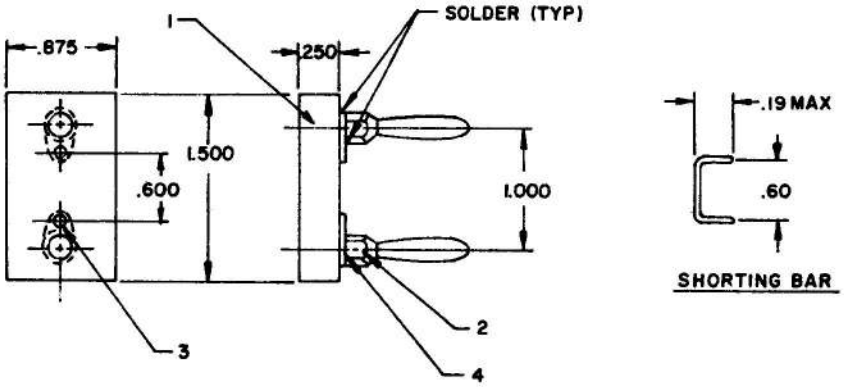
Part or Identifying Number (PIN): MS90546 - (dash number from table I).

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TABLE 1. Electrical characteristics (initial).

Dash number MS90546	Former MS part number	Tuning range						SRF (min) at specified inductance	DC Resistance (max) (0hms)	Rated DC current (mA)
		Inductance (min)	Q (min)	Test frequency	Inductance (max)	Q (min)	Test frequency			
-32	MS90546-01	.23µH	90	25 MHz	.40 µH	100	25 MHz	210 MHz	.05	1800
-33	MS90546-02	.29µH	95	25 MHz	.64 µH	110	25 MHz	165 MHz	.055	1720
-34	MS90546-03	.37µH	100	25 MHz	.90 µH	110	25 MHz	140 MHz	.06	1650
-35	MS90546-04	.50µH	105	25 MHz	1.20 µH	95	7.9 MHz	125 MHz	.08	1430
-36	MS90546-05	.78µH	105	25 MHz	1.80 µH	90	7.9 MHz	95 MHz	.09	1340
-37	MS90546-06	1.10µH	75	7.9 MHz	2.50 µH	65	7.9 MHz	75 MHz	.10	1280
-38	MS90546-07	1.30µH	65	7.9 MHz	3.60 µH	65	7.9 MHz	68 MHz	.23	840
-39	MS90546-08	2.00µH	70	7.9 MHz	6.00 µH	75	7.9 MHz	56 MHz	.35	680
-40	MS90546-09	3.00µH	80	7.9 MHz	9.00 µH	100	7.9 MHz	45 MHz	.75	465
-41	MS90546-10	4.00µH	80	7.9 MHz	11.00 µH	90	2.5 MHz	39 MHz	1.00	405
-42	MS90546-11	7.50µH	80	7.9 MHz	18.00 µH	95	2.5 MHz	32 MHz	1.40	340
-43	MS90546-12	12.00µH	70	2.5 MHz	30.00 µH	100	2.5 MHz	21 MHz	1.50	315
-44	MS90546-13	15.00µH	70	2.5 MHz	44.00 µH	90	2.5 MHz	16 MHz	2.00	274
-45	MS90546-14	25.00µH	75	2.5 MHz	65.00 µH	90	2.5 MHz	13 MHz	2.20	261
-46	MS90546-15	35.00µH	80	2.5 MHz	90.00 µH	85	2.5 MHz	10 MHz	3.00	223
-47	MS90546-16	55.00µH	80	2.5 MHz	130.00 µH	85	790 kHz	8 MHz	3.80	198
-48	MS90546-17	80.00µH	70	2.5 MHz	200.00 µH	85	790 kHz	6.8 MHz	4.20	189
-49	MS90546-18	120.00µH	70	790 kHz	290.00 µH	85	790 kHz	5.6 MHz	4.50	182
-50	MS90546-19	200.00µH	60	790 kHz	430.00 µH	75	790 kHz	4.5 MHz	6.00	158
-51	MS90546-20	300.00µH	50	790 kHz	620.00 µH	70	790 kHz	3.2 MHz	13.00	107
-52	MS90546-21	400.00µH	50	790 kHz	1.00 mH	60	790 kHz	2.4 MHz	17.00	94
-53	MS90546-22	600.00µH	35	790 kHz	1.70 mH	50	250 kHz	1.9 MHz	19.00	82
-54	MS90546-23	1.05 mH	35	250 kHz	2.70 mH	50	250 kHz	1.28 MHz	23.00	74
-55	MS90546-24	1.70 mH	30	250 kHz	4.00 mH	40	250 kHz	1.10 MHz	33.00	62
-56	MS90546-25	2.50 mH	30	250 kHz	6.00 mH	35	250 kHz	840 kHz	41.00	56
-57	MS90546-26	3.70 mH	30	250 kHz	9.00 mH	35	250 kHz	700 kHz	61.00	46
-58	MS90546-27	5.80 mH	30	250 kHz	13.00 mH	30	79 kHz	600 kHz	81.00	40
-59	MS90546-28	8.50 mH	30	250 kHz	22.00 mH	35	79 kHz	370 kHz	120.00	32
-60	MS90546-29	13.00 mH	25	79 kHz	30.00 mH	35	79 kHz	280 kHz	170.00	27
-61	MS90546-30	21.00 mH	20	79 kHz	47.00 mH	25	79 kHz	210 kHz	230.00	23
-62	MS90546-31	35.00 mH	20	79 kHz	70.00 mH	25	79 kHz	140 kHz	345.00	19

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Item	Description	Qty
1	Lucite block	1
2	#103 banana plug or equivalent	2
3	Nugent PS-402-60 or equal	2
4	Copper lug	2

INCHES	mm
.19	4.83
.250	6.35
.600	15.24
.875	22.23
1.000	25.40
1.500	38.10

FIGURE 2. Test fixture for electrical measurements.

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from Initial measurement		Allowable percent from specified minimum value in electrical characteristics (initial) table	
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±2	---	---	-10
Group III	±5	±(2% +.001 ohm)	-5	-15
Group IV	±3	±(2% +.001 ohm)	-5	-10
Conformance inspection group C				
Subgroup I	±2	---	---	-10
Subgroup II	±3	±(2% +.001 ohm)	-5	-10
Subgroup III	±5	±(2% +.001 ohm)	-5	-10

Application notes:

1. Core shall be set at the maximum specified inductance value indicated in the electrical characteristics (initial) table prior to all inspection tests. This core setting shall not be changed until electrical characteristics (final) measurements are performed.
2. Temperature rise and terminal pull tests are applicable.
3. Barometric pressure (test condition C) is applicable.
4. Operating torque shall be maximum of 6 ounce-inches and a minimum of 0.5 ounce-inch for tuning coil.
5. The test fixture in the diagram above shall be used for electrical measurements. The effective inductance shall be determined for all coils whenever this measurement is specified. Allowance of approximately .03 μH shall be determined for internal inductance of the Q-meter and test fixture when measuring inductance values of 10 μH or less. The shorting bar shall be made of solid copper wire (AWG #20) bent to conform to the body configuration of the coil under test (see diagram above).

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Referenced documents. In addition to MIL-PRF-15305 this document references:

MIL-STD-202-105 MIL-STD-202-213 MIL-STD-202-301 ASTM B227

Custodians:
Army – CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA – CC

Project 5950-2021-009

Review activities:
Army – AT, CR4, MI
Navy – AS, CG, MC, OS, SH

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